

SODANKYLA KOKKA, Finland

WMO: 027190

Lat: 67.821N

Lon: 27.746E

Elev: 240

StdP: 98.47

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.9	(c) -30.0	(d) -36.3	(e) 0.1	(f) -33.0	(g) -33.1	(h) 0.2	(i) -30.0	(j) 11.4	(k) -7.9	(l) 10.3	(m) -7.5	(n) 1.5	(o) 190	(p) 0.771

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 24.8	(d) 16.8	(e) 22.4	(f) 15.5	(g) 20.2	(h) 14.6	(i) 18.0	(j) 22.6	(k) 16.7	(l) 20.9	(m) 15.6	(n) 19.2	(o) 3.9	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.4	(b) 12.0	(c) 19.7	(d) 15.1	(e) 11.0	(f) 18.2	(g) 14.0	(h) 10.3	(i) 17.2	(j) 51.8	(k) 22.6	(l) 47.9	(m) 21.2	(n) 44.6	(o) 19.2	(p) 21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.8	7.7	-37.5	26.3	2.0	2.3	-39.0	28.0	-40.2	29.3	-41.3	30.6	-42.8	32.3
			-37.3	19.6	2.1	1.8	-38.8	20.9	-40.1	21.9	-41.3	22.9	-42.8	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-14.0	-12.8	-9.2	-1.9	4.9	10.1	13.9	11.2	7.0	-0.6	-6.3	-10.6
	DBStd	11.03	7.10	8.75	6.70	4.74	4.69	4.06	3.56	3.42	3.22	4.47	6.20	7.10
	HDD10.0	4132	743	637	594	357	168	47	7	26	100	328	488	637
	HDD18.3	6918	1001	871	852	607	416	249	141	221	341	586	738	896
	CDD10.0	265	0	0	0	0	12	50	129	65	9	0	0	0
	CDD18.3	9	0	0	0	0	0	2	6	1	0	0	0	0
	CDH23.3	130	0	0	0	0	3	27	81	19	0	0	0	0
	CDH26.7	17	0	0	0	0	0	2	11	4	0	0	0	0
Wind	WSAvg	3.5	3.1	3.4	3.8	3.4	3.6	4.2	3.6	3.5	3.6	3.6	3.2	3.2
Precipitation	PrecAvg	575	37	34	31	33	44	62	74	62	54	52	48	42
	PrecMax	722	71	76	72	87	85	107	133	125	117	106	79	71
	PrecMin	447	16	15	8	13	14	17	23	21	7	18	14	10
	PrecStd	76	14	17	15	16	17	24	27	26	26	22	18	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	2.5	4.5	11.3	22.7	26.7	28.2	26.3	17.7	8.9	4.8	1.5
		MCWB	-0.5	1.6	2.0	6.1	12.7	16.4	17.7	18.9	13.2	7.9	3.9	1.1
	2%	DB	-1.6	0.6	2.7	7.5	19.5	22.6	25.5	21.9	15.7	7.8	2.7	0.4
		MCWB	-1.8	0.1	0.6	3.5	11.3	15.2	17.1	16.5	12.0	6.8	2.3	0.2
	5%	DB	-2.9	-0.2	1.1	5.6	15.7	19.7	23.6	19.5	14.1	6.5	1.4	-0.2
		MCWB	-3.1	-0.5	-0.3	2.5	9.4	13.6	16.5	14.8	11.2	5.7	1.2	-0.4
	10%	DB	-4.4	-1.5	0.0	4.0	12.8	17.3	21.4	17.6	12.5	5.2	0.6	-1.4
		MCWB	-4.6	-1.9	-0.9	1.6	8.2	12.1	15.5	13.8	10.7	4.6	0.4	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	1.8	2.2	6.3	13.5	18.0	20.2	19.5	14.2	8.8	4.6	1.2
		MCDB	-0.3	2.4	4.1	11.3	20.6	24.2	23.9	25.0	16.4	9.0	4.8	1.4
	2%	WB	-1.8	0.4	1.0	4.0	11.8	16.4	18.5	17.1	12.8	6.8	2.3	0.3
		MCDB	-1.6	0.7	2.3	6.7	18.1	21.7	23.1	21.0	14.9	7.4	2.5	0.4
	5%	WB	-3.0	-0.6	0.1	2.9	10.3	14.7	17.5	15.6	11.8	5.7	1.2	-0.3
		MCDB	-2.9	-0.3	0.9	5.0	15.4	18.1	21.9	18.3	13.4	6.4	1.4	-0.1
	10%	WB	-4.6	-1.9	-0.9	2.0	8.5	13.0	16.4	14.4	10.7	4.6	0.5	-1.7
		MCDB	-4.4	-1.6	-0.1	3.5	12.1	16.6	20.3	16.8	12.2	5.1	0.6	-1.5
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.7	11.5	9.2	9.1	8.5	9.6	9.2	7.2	5.2	6.0	8.2
		MCDBR	8.8	6.0	8.3	9.9	14.8	13.0	14.4	13.0	10.8	5.5	4.4	5.3
		MCWBR	8.8	5.7	7.0	6.5	7.8	6.3	7.3	7.6	7.3	4.7	4.4	5.3
	5% WB	MCDBR	9.0	6.1	6.6	8.6	14.0	10.6	12.4	11.2	8.7	4.9	4.2	5.3
		MCWBR	9.0	5.8	5.8	5.8	7.9	5.9	7.0	7.3	6.3	4.5	4.2	5.4
Clear-Sky Solar Irradiance	taub		0.193	0.253	0.291	0.321	0.340	0.341	0.354	0.335	0.308	0.272	0.355	N/A
	taud		1.916	2.077	2.147	2.198	2.393	2.471	2.456	2.482	2.536	2.300	2.729	N/A
	Ebn at Noon		113	544	739	823	854	863	837	814	752	537	156	0
	Edh at Noon		10	49	82	104	97	93	91	80	59	39	13	0
All-Sky Solar Radiation	RadAvg		0.04	0.52	2.01	3.83	4.71	5.00	4.56	3.30	1.73	0.56	0.10	0.00
	RadStd		0.00	0.06	0.14	0.24	0.45	0.42	0.48	0.31	0.17	0.10	0.01	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page